

# SOMIL GUPTA

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## SUMMARY

Computer Science undergraduate with hands-on experience in backend development, systems programming, and API design using Node.js, NestJS, Rust, and C++. Published open-source packages and contributed to production systems.

## EDUCATION

**Indian Institute of Information Technology Bhagalpur**

May 2023 – Present

Bachelor of Technology in Computer Science and Engineering — CGPA: 8.04/10

## EXPERIENCE

**Recrivio**

May 2026 – Present

Software Development Engineer Intern

Onsite

- Architected **Recrith**, a background verification system, implementing role-based access control (RBAC) via **NestJS** to handle authorization for **100%** of incoming verification requests.
- Built an in-house OCR engine using **ONNX models** and **OpenCV** for identity verification and automated data masking, eliminating third-party API dependency and cutting operational costs by **30%**; also integrated automated multi-language transcription via **FFmpeg** and scheduled cron workers.

**Creova**

May 2025 – Jul 2025

Software Development Engineer Intern

Remote

- Collaborated with a cross-functional team to design and deploy **3 production-grade full-stack applications** using **Next.js**, **Node.js**, and **MongoDB**, improving scalability and reducing page load time by **35%** for **10K+** users.
- Revamped **8 Redis-cached RESTful APIs** and migrated **2.5K+** lines of legacy JavaScript to TypeScript, cutting response latency by **40%** and lowering production bug frequency by **60%**.

## TECHNICAL SKILLS

**Languages:** Rust, C++, C, Python, JavaScript, TypeScript, SQL, Bash, Golang

**Frontend:** React.js, Next.js, Tailwind CSS, shadcn/ui, Framer Motion

**Backend:** Node.js, Express.js, NestJS, FastAPI, Axum, REST APIs, WebSockets

**Databases and Cloud:** PostgreSQL, MongoDB, Redis, Prisma, Drizzle ORM, Docker, AWS (EC2, S3), CI/CD

**AI and GenAI:** LangChain, ONNX Models, OpenCV, OpenAI APIs, Prompt Engineering

## PROJECTS

**Lithos** | C++17, Rust, Next.js, Axum

Jun 2026 – Jul 2026

- Designed an embeddable Log-Structured Merge (LSM) Tree key-value database engine in **C++17** featuring Skip Lists, Write-Ahead Logs (WAL), and Bloom Filters for high-throughput data operations.
- Exposed the storage engine through a **Rust** cxx FFI bridge and Axum REST API, achieving an estimated throughput of **384K req/s** (**13x faster** than comparable Node.js implementations) while minimizing P99 FFI latency overhead.

**Impulse** | Next.js, TypeScript, PostgreSQL, Prisma, Docker

Jul 2025 – Sep 2025

- Launched an API testing tool for HTTP workflows, achieving **99.9% authentication reliability** and **35% faster** database query responses through Prisma index optimization.

## ACHIEVEMENTS AND OPEN SOURCE

'guts-validator' (NPM Package), Jun-Jul 2026 – Engineered a single-compilation schema validator in native Rust using zero-copy architecture, outperforming JS alternatives like Zod.

Open Source Contributor, Requestly (YC '24), Aug-Oct 2025 – Contributed to the core repository, enhancing API testing workflow features.

NMIT Hacks 2025 – 1<sup>st</sup> Place, May 2025 (50K+ participants) – Built an AI medical diagnostic system using CNNs and multi-modal imaging in 48 hours.

Smart India Hackathon 2024 – Grand Finalist, Dec 2024 (Top 5/3K+ teams) – Devised a Deep Q-Learning traffic optimization agent with TensorFlow and SUMO, cutting wait time by **30%**.

Competitive Programming – LeetCode: 1710 rating (Top 10%), 500+ problems solved. CodeChef 3-Star: 1710 rating.

## CERTIFICATIONS

**ServiceNow Certified System Administrator (CSA)**

Jun 2026

ServiceNow

**Introduction to Cybersecurity**

Mar 2025

Cisco